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IFAC PapersOnLine 55-1 (2022) 235–240

# Extended Kalman filter-based precise orbit estimation of LEO satellites using GPS range measurements

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**Abstract:** In order to meet the accuracy requirements of scientific mission applications of Low Earth Orbit (LEO) satellites, precise orbit solution with at least less than a meter accuracy at a fast update rate is required. To cater to these needs, this paper proposes a methodology for fast and precise estimation on ground, of orbits (position and velocity) of LEO satellites, based on Extended Kalman Filter (EKF) estimation technique using Global Positioning System (GPS) precise data products. For the proposed Precise Orbit Determination (POD) of LEO satellites, dual frequency code/carrier phase and single frequency code and carrier phase based GPS one-way range measurements are used. All the perturbation forces acting on a LEO spacecraft are modelled, and the errors in GPS measurements are modelled and removed. The proposed methodology when applied to the GPS receiver data of LEO satellites like GRACE, COSMIC, HYSIS, CARTO and MICROSAT resulted in cm level accuracy in position and mm/s level accuracy in velocity.

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*Keywords:* Extended Kalman Filter, Orbit Estimation, LEO satellites, Global Positioning System (GPS), Precise Orbit Determination

## 1. INTRODUCTION

Over the last 30 years, a large number of LEO satellites with GPS receivers have been launched into space, and for several LEO spacecraft missions, GPS-based POD became the primary and fundamental scheme for orbit determination (Yang et al., 2015). Continuous availability, global coverage, accuracy, and three-dimensional nature of GPS measurements make it a more preferred choice for POD compared to other tracking techniques like DORIS and LASER ranging (Montenbruck et al., 2005).

POD techniques for LEO satellites using GPS range measurements fall in three categories: kinematic precise orbit determination (KPOD), dynamic precise orbit determination (DPOD), and reduced dynamic precise orbit determination (RPOD). Among these, kinematic POD requires only spaceborne GPS receiver measurements for LEO spacecraft positioning (JianCheng et al., 2010), and no force modeling and a priori knowledge of spacecraft dynamics are needed. However, kinematic methods are highly sensitive to erroneous range measurements, minimum number of good measurements required (at least four), unfavorable viewing geometry, and measurement outages, which restricts their application in LEO POD. Dynamic orbit determination, in contrast, makes use of known physical models of the spacecraft motion to obtain better position estimates (Yang et al., 2015). The use of accurate models also allows an averaging of measurements from different epochs and the trajectory can even be propagated across data gaps (Swatschina, 2012). However, the computational effort involved is high prohibiting its onboard implementation. Hence, the concept of reduced dynamic orbit determination has been proposed (Montenbruck et al., 2005).

In RPOD, force model used is not accurate, and therefore, unmodelled errors in the dynamics of spacecraft is accounted for by estimating empirical accelerations that are adjusted along with other parameters in the orbit determination process. RPOD technique is preferred for onboard LEO orbit estimation as it fuses the advantages of both kinematic positioning and fully dynamic approach, with reduced computational effort compared to full dynamic modelling of orbital forces (Montenbruck et al., 2005).

If the scientific application that requires POD is onboard based, RPOD is preferred due to reduced computational effort compared to DPOD as it provides faster solution with slight trade off in accuracy. For applications which does not require real time updates, DPOD is preferred over kinematic and reduced dynamic method. DPOD uses full force dynamic model in orbit estimation process by considering all the forces acting on the spacecraft in Earth Centered Inertial (ECI) frame. With precise GPS ephemeris and clock products, highly precise LEO orbit can be achieved via dynamic precise orbit determination methodology. Hence, accurate modelling needs to be done of all the forces acting on LEO satellite to employ DPOD.

Major challenges in precise orbit estimation of LEO satellites are precise modelling of GPS range measurements (discussed in Section 2.2) by removing all the errors in line

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Peer review under responsibility of International Federation of Automatic Control.
10.1016/j.ifacol.2022.04.039

of sight, accurate computation of all forces acting on LEO spacecraft (orbital dynamics) as well as determination of appropriate measurement combinations (Gruber, 2005), differencing and estimation techniques to be used.

After completing data processing and range modelling, error-free measurements are used for estimation of LEO satellite orbit. Estimation technique used for LEO POD can either be sequential or batch. In this paper, EKF based estimation is chosen over Batch Least Squares based estimation for faster and instantaneous update of the solution. Montenbruck et al. (2005) and Swatschina (2012) talks about the cm-level accuracy of the LEO orbit solution that can be achieved using GPS range measurements. However, reproducing the accuracy levels that they claim is hard to achieve as many details in range and orbit modelling are missing. So, in this paper, we provide information about all the input and the details of the models used that will provide cm-level orbit accuracy. In order to get precise estimate of LEO satellite orbit, DPOD technique is chosen in orbit modelling. While doing the work presented in this paper, we explored various combinations of models and what we found as combination that works best is presented.

This paper lays down the complete methodology for achieving POD of LEO satellites and demonstrates it using real satellite data. Section 2 contains complete range modelling of the GPS range measurements for P3, L3 and single frequency ionospheric error free (P1+L1)/2 (GRAPHIC) combinations and also explains the several orbit models incorporated to obtain the forces and hence the accelerations acting on the LEO satellite. EKF algorithm adopted for precise estimation, the numerical integration scheme used, the inputs required and data processing of range measurements are explained in Section 3. Results and analysis done for various LEO satellites are provided in Section 4. Section 5 summarizes the discussion of orbit estimation results and states the conclusions.

## 2. METHODOLOGY

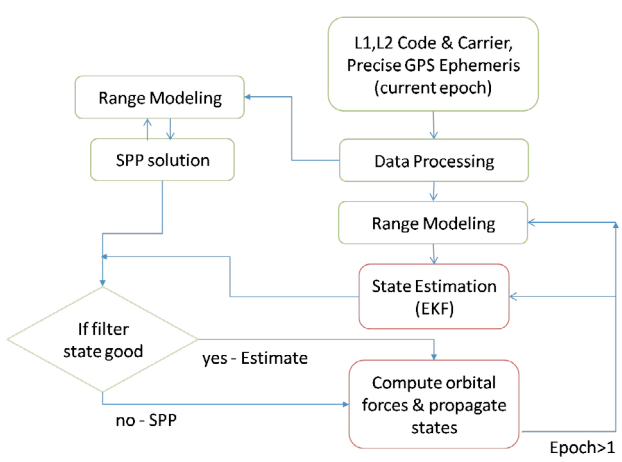


Fig. 1. Process flow diagram of methodology

The methodology followed in this paper for precise orbit determination of LEO satellites using GPS measurements is illustrated in Fig. 1. GPS carrier phase range measurements are first processed to remove the outliers and cycle slips and are smoothed. All the line of sight errors in the range are modelled and removed. Single point positioning (SPP) is used to get the initial estimate of LEO satellite orbit, which will serve to initialize the Extended Kalman Filter (EKF) algorithm. We chose EKF over other estimation techniques owing to the lower convergence time (less than 5 minutes with SPP initial guess) in spite of a slight reduction in accuracy due to the involved linearization of propagation and measurement models. Range modelling and SPP are done in iterative way, to get the final converged SPP orbital position, every epoch. Orbit modelling is used in the time update step of EKF algorithm, where all the forces acting on the LEO satellite are precisely modelled. The processed and corrected measurements are fused with the a priori estimates from the EKF time update step to obtain a posteriori estimates of the orbit. Estimation has been done at an interval with best smooth measurements. Validation of filter estimated states is done using SPP as well as range residue monitoring. In case of filter divergence and covariance going out of bounds, the SPP solution will be used as the estimate for the current time step and filter will be reinitialized with initial covariance values. A priori estimates from EKF for the next time step are also used in range modeling to remove the errors from the range measurements before both of them are fused together. As the filter settles, orbit position accuracy will start improving which in turn improves the range modelling. This process is repeated every epoch to get smooth, precise, and instantaneous LEO orbit solution.

### 2.1 Ionosphere-free range combinations

The passage of the GPS signal through the atmosphere introduces a delay in its reception. This delay needs to be modelled and removed for an accurate range estimate using GPS measurements. The signal from a GPS satellite to a LEO receiver will pass through the troposphere only when the elevation angle between the GPS transmitter and the LEO receiver is less than 5 degrees (Hwang and Tseng, 2009), and the signal delay due to tropospheric effect is considered only in such cases. However, signal delay due to ionosphere has to be accounted for. Several combinations of range measurements can be made to eliminate the delay due to ionosphere. All of them are based on the principle that ionospheric delay is inversely proportional to square of the transmitted frequency. The combinations used for POD in this paper are $P_3$ (ionospheric error free dual frequency code combination), $L_3$ (ionospheric error free dual frequency carrier phase combination), GRAPHIC (group and phase ionospheric error free single frequency code and carrier phase combination) (Montenbruck et al., 2005). These combinations are formed from the code ($P$) and carrier phase ($L$) measurements as

$$P_3 = \frac{1}{(f_1^2 - f_2^2)}(f_1^2 P_1 - f_2^2 P_2) \tag{1}$$

$$L_3 = \frac{1}{(f_1^2 - f_2^2)}(f_1^2 L_1 - f_2^2 L_2) \tag{2}$$

$$GRAPHIC = \frac{P_1 + L_1}{2} \tag{3}$$

where, $f_1$ (1575.42 MHz) and $f_2$ (1227.60 MHz) are GPS carrier wave frequencies in which signals are transmitted, $(P_1, P_2)$ and $(L_1, L_2)$ are code and carrier phase measurements transmitted respectively in frequencies $f_1$ and $f_2$.

### 2.2 Range Modelling

Several errors form part of GPS one-way range measurements which has to be accurately modelled to construct true range from measured pseudo-range (Garcia-Fernandez and Montenbruck, 2005). Range measurement errors include relativistic errors due to earth and satellite motion, Sagnac effect (Larson et al., 2009) due to bending of signal from GPS to LEO satellite, antenna phase centre offsets (PCO) (Jäggi et al., 2009), satellite and receiver clock offset, drift and line of sight integer ambiguities. These can be modelled as follows:

(1) GPS satellite clock offset:
If $\delta t_{\mathrm{s}}$ is the GPS satellite clock offset from broadcast navigation data in seconds, then the error in range measurement is $\delta t_{\mathrm{s}} c$, where $c$ is speed of light.

(2) Sagnac effect (Saphiro delay):
The change in pseudo-range measurement due to Sagnac effect can be modelled as

$$\delta\rho_{\mathrm{sag}} = \frac{(\boldsymbol{r}_{\mathrm{s}} - \boldsymbol{r}_{\mathrm{r}}) \cdot \boldsymbol{v}_{\mathrm{s}}}{c} \tag{4}$$

where $\boldsymbol{r}_{\mathrm{r}}$ is the LEO satellite position vector, and $\boldsymbol{r}_{\mathrm{s}}$ and $\boldsymbol{v}_{\mathrm{s}}$ are the position and velocity vectors of the GPS satellite, all expressed in ECI frame.

(3) Relativistic effects (bending of the signal):
The correction to pseudo-range measurement owing to relativistic effect is given as

$$\delta\rho_{\mathrm{rel}} = 2\left(\frac{\boldsymbol{r}_{\mathrm{s}} \cdot \boldsymbol{v}_{\mathrm{s}}}{c} - \frac{\boldsymbol{r}_{\mathrm{r}} \cdot \boldsymbol{v}_{\mathrm{r}}}{c}\right) \tag{5}$$

(4) Antenna error:
PCOs of the GPS and LEO satellites' antennas causes an error. The error introduced in the range measurement due to this can be modeled as

$$\delta\rho_{\mathrm{ant}} = (\boldsymbol{r}_{\mathrm{r}} - \boldsymbol{r}_{\mathrm{s}}) \cdot \boldsymbol{\delta}_{\mathrm{ant_I}} \tag{6}$$

where $\boldsymbol{\delta}_{\mathrm{ant_I}}$ is the PCO vector represented in ECI frame. Ground calibrated PCO vector values in body frame for LEO satellites are available in CDAAC website. This vector needs to be transformed in ECI frame, in order to compute the line of sight phase centre offset values.

*Computed Range:* Accounting for the above range measurement errors, the computed range (time of flight $\times$ $c$) for each line of sight can be expressed as

$$\rho_{\mathrm{c}} = \rho_{\mathrm{g}} - \delta t_{\mathrm{s}} c - \delta\rho_{\mathrm{sag}} + \delta\rho_{\mathrm{rel}} + \delta\rho_{\mathrm{ant}} + b_0 - N \tag{7}$$

where $\rho_{\mathrm{g}}$ is the actual geometric distance between the receiver and the GPS satellite, $N$ is the integer ambiguity associated with carrier phase (wavelength multiplied by initial number of cycles missed) in meters, and $b_0$ is the LEO satellite's receiver clock bias in meters. Integer ambiguities (corresponding to each line of sight/ GPS satellite) and receiver clock bias forms part of the estimated parameters. Receiver clock estimation or modelling plays an important role in precise orbit determination of LEO satellites (Weinbach and Schon, 2012). Time varying receiver clock bias is modeled as (DongJu and Bin, 2007)

$$b_0(t+h) = b_0(t) + b_1 h \tag{8}$$

where $b_1$ is the receiver clock drift in m/s, and $h$ is the time interval between two consecutive GPS measurements.

### 2.3 Single Point Positioning (SPP)

SPP uses trilateration technique to fix the position of a satellite. The basic idea is that, if four or more GPS satellites are visible to the LEO satellite, then the 4 unknowns - 3 position values $(x_{\mathrm{r}}, y_{\mathrm{r}}, z_{\mathrm{r}})$ and 1 receiver clock bias $(b_0)$ - can be solved by least squares approach. The parameters $\boldsymbol{X} = [x_{\mathrm{r}}\, y_{\mathrm{r}}\, z_{\mathrm{r}}\, b_0]^T$ are estimated from the observed GPS range measurement vector $\boldsymbol{Z} = [\rho_1\, \rho_2\, \cdots\, \rho_m]^T$ where $\rho_i$, $i = 1, \ldots, m$ is the observed range (after removing all the errors except receiver clock bias using range modelling) from $i^{\mathrm{th}}$ GPS satellite to receiver.

SPP involves an iterative procedure, and we implemented the algorithm given in Sickle (2021). The position estimate from SPP is noisy and erroneous and will have an average position accuracy of 5 to 10 meters. Nonetheless, the SPP estimate can be used as an initial guess for better estimation techniques.

## 3. ESTIMATION TECHNIQUE

### 3.1 Orbit Modelling

Equation of perturbation forces acting on LEO satellite is

$$\begin{aligned}\boldsymbol{a} =& \boldsymbol{a}_{\mathrm{g}} + \boldsymbol{a}_{\mathrm{asp}} + \boldsymbol{a}_{\mathrm{sun}} + \boldsymbol{a}_{\mathrm{moon}} + \boldsymbol{a}_{\mathrm{srp}} \\ &+ \boldsymbol{a}_{\mathrm{tide}} + \boldsymbol{a}_{\mathrm{rel}} + \boldsymbol{a}_{\mathrm{drag}}\end{aligned} \tag{9}$$

where the terms on the right hand side are accelerations due to Earth's gravitational pull and its oblateness, gravitational attraction of Sun and Moon (3rd body pull), solar radiation pressure, solid and ocean tides, relativistic effect and atmospheric drag, respectively. The modeling of each term on the right hand side of Eq. (9) is done as in Montenbruck and Ramos-Bosch (2007).

*Earth's oblateness and tides:* After several trials, in this paper, EGM-2008 dynamical model with $120 \times 120$ order and degree are selected for Earth's asphericity modelling for better accuracy. Tidal deformations of the Earth cause an acceleration of about $200\,\mathrm{nm/s^2}$ at 400 km altitude and a simple approximation for the $k_2$ solid-Earth tide has therefore been considered.

*Third-body perturbations:* Gravitational perturbations of the Sun and Moon amount to accelerations of the order of 300 to $1000\,\mathrm{nm/s^2}$ for satellites in the immediate neighbourhood of Earth. Hence, luni-solar coordinates need not be known with utmost precision to meet the targeted overall modelling accuracy. For the present purpose, short series expansions with representative accuracies of 0.3 to 1.5 mrad have been adopted that induce only a negligible error into the acceleration modelling (Montenbruck et al., 2005). Aside from the modelling of third-body accelerations, the luni-solar coordinates are also used for evaluating the solid Earth tides and the solar-radiation pressure.

*Solar Radiation Pressure (SRP):* The acceleration of a spacecraft with coefficient of reflectivity $C_r$ exposed to irradiance $I_{\mathrm{sat}}$ from sun is given as

$$\boldsymbol{a}_{\mathrm{srp}} = \frac{C_{\mathrm{r}} S_{\mathrm{shadow}} I_{\mathrm{sat}}}{c} \hat{\boldsymbol{R}}_{\mathrm{satsun}} \tag{10}$$

where $\hat{\boldsymbol{R}}_{\mathrm{satsun}}$ is the unit vector from Sun to the satellite, and $S_{\mathrm{shadow}}$ takes value between 0 and 1 depending on Earth's shadow on the satellite being umbra or penumbra or fully lit. Varying illumination conditions along the satellite orbit are accounted for by a cylindrical shadow

model (Montenbruck et al., 2005). Irradiance at the satellite surface due to Sun can be computed using irradiance value at Earth's surface (which is known) (Montenbruck and Ramos-Bosch, 2007).

*Relativistic effects:* Acceleration due to general relativity can be modelled and separated from purely Newtonian acceleration. We have used the relativistic correction to acceleration of a satellite in earth's orbit using earth's angular momentum as given in McCarthy and Petit (2003).

*Atmospheric drag:* As a final constituent, the dynamical model accounts for atmospheric drag – a major source of orbital perturbations at low altitudes perturbations. MSIS model has been selected over Harris-Priester (atmospheric density based on mean solar activity) for modelling the density of the upper atmosphere with better accuracy. Acceleration due to atmospheric drag is given as

$$\boldsymbol{a}_{\text{drag}} = -\frac{1}{2} C_d \rho v^2 \hat{\boldsymbol{v}} \tag{11}$$

where $C_d$ is the ballistic drag coefficient describing interaction of atmospheric element with satellite's surface material, $\rho$ is the atmospheric density at the satellite's orbit, which is derived using MSIS model (Picone et al., 2002), $v$ is the satellite's speed, and $\hat{\boldsymbol{v}}$ is the unit vector in the direction of the satellite's velocity.

Once the total acceleration is computed, it is integrated to get the velocity and position of the LEO satellite.

### 3.2 Extended Kalman Filter

We use an Extended Kalman filter setting to estimate the following parameters: LEO satellite's position and velocity in ECI frame – $(x, y, z)$, $(v_x, v_y, v_z)$, coefficient of reflectivity (solar radiation pressure coefficient) – $(C_r)$, ballistic (drag) coefficient – $(C_d)$, LEO satellite's clock bias and drift – $(b_0, b_1)$, and integer ambiguity (at every line of sight) – $N_i$, $i = 1, \ldots, m$ where $m$ is the number of GPS measurements. The maximum value $m$ can take is 32 as there are 32 GPS satellites. Therefore, the maximum number of states can be 42 (10+32). Thus, the state vector to be estimated is given as

$$\boldsymbol{X} = \begin{bmatrix} x, & y, & z, & v_x, & v_y, & v_z, & b_0, & b_1, & C_r, & C_d, \\ N_1, & N_2, & \ldots & N_m & & & & & & \end{bmatrix}^T$$

Let $\boldsymbol{Z} = [\rho_{T_1} \ \rho_{T_2} \ \cdots \ \rho_{T_m}]^T$ be the vector of observed range measurements from $m$ GPS satellites, where $\rho_{T_i}$ is the smoothed GPS carrier phase range measurement from $i^{\text{th}}$ GPS satellite which is ionospheric error, outlier, and cycle-slips-free. The measurement model is

$$\boldsymbol{Z} = \boldsymbol{Z}_c + \boldsymbol{w} \tag{12}$$

with $\boldsymbol{Z}_c = [\rho_c^1 \ \rho_c^2 \ \cdots \ \rho_c^m]^T$ where $\rho_c^i$ is the computed range constructed from estimated state parameters as in Eq. (7), and $\boldsymbol{w}$ is the measurement noise with covariance $\mathbf{R}$.

A priori estimate of $\boldsymbol{X}$ and its covariance are found using time update equations while the posteriori estimate and the corresponding covariance matrix are found using measurement update equations. Measurement model/sensor dynamics matrix $\mathbf{H}$ is Jacobian of measurement with respect to state parameters. If $m$ GPS satellites measurements are available at a particular instant, then corresponding $\mathbf{H}$ matrix is computed as,

$$\mathbf{H} = \begin{bmatrix} \frac{\partial \rho_c^1}{\partial x_r} & \frac{\partial \rho_c^1}{\partial y_r} & \frac{\partial \rho_c^1}{\partial z_r} & \frac{\partial \rho_c^1}{\partial v_{x_r}} & \cdots & \frac{\partial \rho_c^1}{\partial C_d} & \frac{\partial \rho_c^1}{\partial N_1} & \cdots & \frac{\partial \rho_c^1}{\partial N_m} \\ \frac{\partial \rho_c^2}{\partial x_r} & \frac{\partial \rho_c^2}{\partial y_r} & \frac{\partial \rho_c^2}{\partial z_r} & \frac{\partial \rho_c^2}{\partial v_{x_r}} & \cdots & \frac{\partial \rho_c^2}{\partial C_d} & \frac{\partial \rho_c^2}{\partial N_1} & \cdots & \frac{\partial \rho_c^1}{\partial N_m} \\ \frac{\partial \rho_c^3}{\partial x_r} & \frac{\partial \rho_c^3}{\partial y_r} & \frac{\partial \rho_c^3}{\partial z_r} & \frac{\partial \rho_c^3}{\partial v_{x_r}} & \cdots & \frac{\partial \rho_c^3}{\partial C_d} & \frac{\partial \rho_c^3}{\partial N_1} & \cdots & \frac{\partial \rho_c^1}{\partial N_m} \\ \vdots & & & & & & & & \\ \frac{\partial \rho_c^m}{\partial x_r} & \frac{\partial \rho_c^m}{\partial y_r} & \frac{\partial \rho_c^m}{\partial z_r} & \frac{\partial \rho_c^m}{\partial v_{x_r}} & \cdots & \frac{\partial \rho_c^m}{\partial C_d} & \frac{\partial \rho_c^m}{\partial N_1} & \cdots & \frac{\partial \rho_c^1}{\partial N_m} \end{bmatrix} \tag{13}$$

Time update:

$$\boldsymbol{X}_{k+1}^- = \boldsymbol{\Phi}_{k+1|k} \boldsymbol{X}_k \tag{14}$$

$$\mathbf{P}_{k+1}^- = \boldsymbol{\Phi}_{k+1|k} \mathbf{P}_k \boldsymbol{\Phi}_{k+1|k}^T + \mathbf{Q} \tag{15}$$

Measurement update:

$$\mathbf{K}_{k+1} = \mathbf{P}_{k+1}^- \mathbf{H}_{k+1}^T [\mathbf{H}_{k+1} \mathbf{P}_{k+1}^- \mathbf{H}_{k+1}^T + \mathbf{R}]^{-1} \tag{16}$$

$$\boldsymbol{X}_{k+1} = \boldsymbol{X}_{k+1}^- + \mathbf{K}_{k+1} \boldsymbol{\Delta Z}_{k+1} \tag{17}$$

$$\mathbf{P}_{k+1} = [\mathbf{I} - \mathbf{K}_{k+1} \mathbf{H}_{k+1}] \mathbf{P}_{k+1}^- \tag{18}$$

where $\boldsymbol{X}_k^-$ and $\boldsymbol{X}_k$ are the a priori and a posteriori estimates, respectively, of $\boldsymbol{X}$ at time step $k$. The state-transition matrix $\boldsymbol{\Phi}_{k+1|k}$ propagates states from time step $k$ to the next time step. The matrices $\mathbf{P}_k^-$ and $\mathbf{P}_k$ are the a priori and a posteriori estimates, respectively, of the state error covariance matrix at time step $k$. At time step $k$, the measurement matrix is $\mathbf{H}_k$ computed as in Eq. (13) and the Kalman gain matrix is $\mathbf{K}_k$. The measurement residue $\boldsymbol{\Delta Z}_k$ is $\boldsymbol{Z} - \boldsymbol{Z}_c$ at time step $k$. The process noise covariance matrix $\mathbf{Q}$ accounts for modelling errors, and $\mathbf{R}$ is the measurement noise covariance matrix.

In order to get the state transition matrix from acceleration partials, Runge-Kutta 4$^{\text{th}}$ order integration scheme is used. The computed state transition matrix is used in Eq. (15) to get the predicted state error covariance matrix. Measurement update step in EKF has been done sequentially at every line of sight to avoid matrix inverse computation (also singularity of the matrix) (Montenbruck and Ramos-Bosch, 2007). The number of range measurements keep on varying depending upon the visibility of GPS satellites. Hence, at every epoch, $m$ value may change, and then the parameter vector $\boldsymbol{X}$ size keeps changing. For every epoch, measurement update step equations are repeated $m$ times, with every step using the updated results from previous step. In case of filter divergence and covariance going out of bounds, filter re-initialization is done using computed SPP solution. For the sake of computational advantage, we trade a bit of accuracy and solve for the integer ambiguities as floating points (not imposing the constraint of them needing to be integers).

All the inputs required for this technique are: GPS C/A code, carrier, Doppler and signal to noise ratio (SNR - $L_1, L_2$ frequencies), GPS precise (navigation) data products, GPS precise ephemeris, clock (navigation data) along with time-stamping, earth orientation parameters, time offsets, leap second, antenna phase centre offsets (PCO), attitude information (for PCO transformation), GPS range measurements data for LEO satellites considered (sampling interval: 1 sec), single point positioning (SPP) solution for initializing filter, configuration parameters (initial settings for consider parameters ($C_r$, $C_d$) and covariances - initial $\mathbf{P}, \mathbf{Q}, \mathbf{R}$). Initial $\mathbf{P}$ value for position

parameters are chosen as 10 and velocity parameters are chosen as 0.5 (based on SPP accuracy). Since full force propagation model is used and the measurement update happens every second, the error due to improper modelling is less than a cm within a second. Each and every force model is independently validated, sensitive analysis is performed to derive the process noise covariance $\mathbf{Q}$. The diagonal entries of $\mathbf{Q}$ corresponding to position parameters are chosen as $10^{-3}$ and those corresponding to velocity are chosen as $10^{-6}$. From the sensitivity studies, it is found that the most critical parameters are perturbation due to Earth's oblateness and atmospheric drag. In case of range modelling, sensitivity analysis is performed on all the errors in the pseudo range. All the line of sight errors are carefully studied and modelled to get the true geometric range. The measurement covariance matrix $\mathbf{R}$ is dynamically chosen for each line of sight GPS measurement, based on elevation angle, with more weightage given for high elevation angle (less noisy measurement at higher elevation angle). As a part of data processing, outlier detection of code range measurements is carried out based on maximum and minimum range rate bounds, maximum and minimum range bounds, and signal to noise ratio (SNR). Cycle-slip detection is carried out using rate of $L_4$ measurements. Smoothing of measurements is done using Hatch filter (smoothing constant = 300 s). $P3$, $L3$, and $GRAPHIC$ combinations are taken for all the individual GPS range measurements continuously at every instant and then smoothed by removing the outliers.

The novelty in the proposed approach consists in using double differencing to remove outliers and cycleslips for both single and dual frequency measurements, dynamically choosing $\mathbf{R}$ based on elevation angle, and applying carrier phase based orbit determination technique for the first time on Indian satellites.

## 4. RESULTS

Table 1. Accuracy of orbit estimation (RSS $3\sigma$) using the proposed approach

| Satellite | $(\mathbf{L_3})$ | | $(\mathbf{P_1}+\mathbf{L_1})/\mathbf{2}$ | | $(\mathbf{P_1})$ | |
|---|---|---|---|---|---|---|
| | Pos. (m) | Vel. (mm/s) | Pos. (m) | Vel. (mm/s) | Pos. (m) | Vel. (mm/s) |
| GRACE (wrt precise) | 1.1 | 3.1 | 1.4 | 3.6 | 3.4 | 4.2 |
| COSMIC (wrt precise) | 0.8 | 2.2 | 1.1 | 2.8 | 3.2 | 3.8 |
| HYSIS (wrt batch) | Not Available | | 1.6 | 3.9 | 3.7 | 5.3 |
| CARTO/ MICROSAT (wrt batch) | Not Available | | Not Available | | 3.5 | 5.7 |

The proposed methodology was applied to single frequency code and carrier phase combination of GPS measurements for 3 days of GRACE-A satellite. The Residual Sum of Squares (RSS) errors in position (Pos.) and velocity (Vel.) when compared against available precise solution (from CDAAC website CDAAC (2020)) are shown in Fig. 2 and Fig. 3, respectively. The position accuracy obtained using our methodology is well within 1.5 m and velocity accuracy is within 3.5 mm/s. Table 1 shows the achieved position and velocity accuracies using the proposed EKF

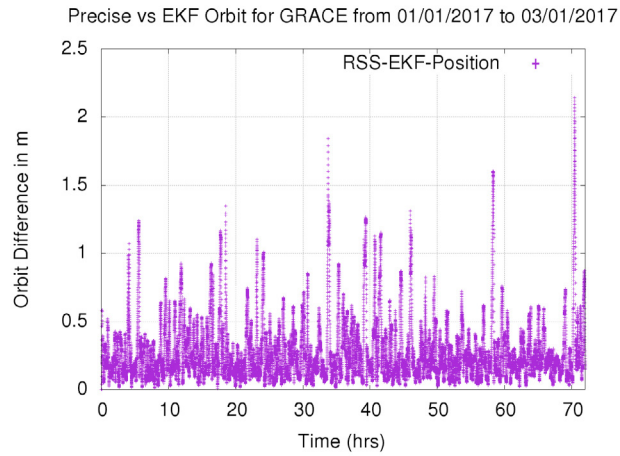


Fig. 2. GRACE-A position error (RSS) as compared with precise

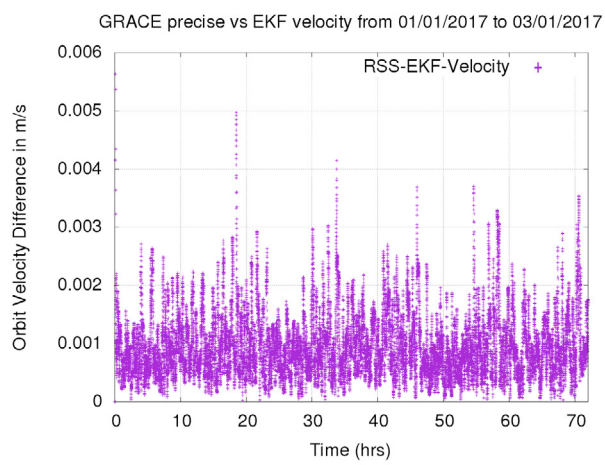


Fig. 3. GRACE-A velocity error (RSS) as compared to precise

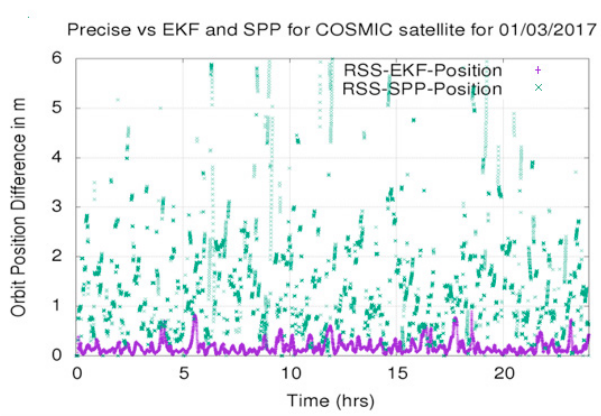


Fig. 4. COSMIC Precise Vs EKF and SPP comparison

estimation for international LEO satellites GRACE and COSMIC, and Indian satellites HYSIS, CARTO and MICROSAT. None of the Indian LEO satellites have dual frequency receivers, and therefore, $L_3$ combination results are not available. Moreover, carrier phase measurements are not available for CARTO and MICROSAT. Estimation results for COSMIC is given in Fig. 4. As can be

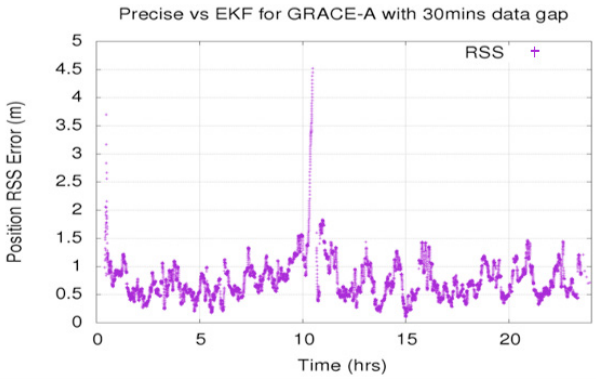


Fig. 5. GRACE-A position error (RSS) with a measurement having 30 min data gap

observed from the figure, the RSS position accuracy of COSMIC satellite using SPP is within 5 m, whereas it is well within 1 m while using EKF. We also used the GPS receiver data of GRACE-A satellite wherein there was a 30 minutes gap. As seen from Fig. 5, which shows RSS error in position when compared against precise solution, in spite of the measurement outage, the position accuracy did not deteriorate beyond 5 m (which is nominal accuracy of SPP solution), as propagation in the absense of measurement is also part of EKF estimation. It can also be noticed from presented result figures that once the filter settles, less than a meter orbit position accuracy is achieved.

## 5. SUMMARY AND CONCLUSIONS

The methodology proposed in this paper ensures fast, continuous, and smooth orbit solution availability even when the number of GPS measurements is less than four (which is the minimum requirement for kinematic SPP solution using trilateration). EKF estimation is preferred over batch least squares estimation owing to faster solution convergence and instantaneous orbit solution availability. The use of EKF algorithm provides better velocity estimate, and hence, better prediction accuracy, thus aiding advance payload operations and planning. While using different combinations of models, we observed that among the orbital models used for LEO satellites, geo-potential and atmospheric drag models form major error sources, and therefore, these need to be modelled more accurately. It is expected that the results obtained using the proposed methodology can be further enhanced up to 30 cm accuracy in orbital position, by fixing the integer ambiguities accurately through decorrelation techniques by exploiting its integer nature, instead of estimating it as a floating point number. It seems worthwhile to explore if other nonlinear estimation techniques like Unscented Kalman Filter (UKF) could further improve the results.

## ACKNOWLEDGEMENTS

The authors acknowledge the useful discussions they had with Dr. Pandiyan.

**Author Contributions:** KS and PM conceived the work, performed the simulations, and prepared the manuscript. MK provided relevant literature references and useful suggestions. JGM did the complete review of the work and technical editing of the paper. NS supported the realization of this work.

## REFERENCES

CDAAC (2020). CDAAC data products. https://cdaac-www.cosmic.ucar.edu/cdaac/products.html. [Online; accessed 15-January-2022].

DongJu, P. and Bin, W. (2007). Zero-difference and single-difference POD for LEO using GPS. In *Chinese Science Bulletin*. August, Vol. 52, No. 15.

Garcia-Fernandez, M. and Montenbruck, O. (2005). LEO satellite navigation errors and VTEC in single-frequency GPS tracking. In *DLR-GSOC*. Radio Science.

Gruber, C. (2005). Outlier detection in CHAMP kinematic orbit data to be used in gravity field determination. In *Institute of Navigation and Satellite Geodesy*. Graz University of Technology, March.

Hwang, C. and Tseng, T.P. (2009). POD for the FORMOSAT-3/COSMIC satellite mission using GPS. In *Journal of Geodesy*.

JianCheng, L., ShouJian, Z., and XianCai, Z. (2010). Precise orbit determination for GRACE with zero-difference kinematic method. In *Chinese Sci. Bull.* Vol.55, No.7.

Jäggi, A., Dach, R., and Montenbruck, O. (2009). Phase centre modelling for LEO GPS receiver antennas and its impact on POD. In *Journal of Geodesy*.

Larson, K.M., Ashby, N., Hackman, C., and Bertiger, W. (2009). An assessment of relativistic effects for low earth orbiters: the GRACE satellites. In *Jet Propulsion Laboratory*. Pasadena, CA.

McCarthy, D.D. and Petit, G. (2003). IERS conventions. In *US Naval Observatory (USNO)*. International Earth Rotation and Reference Systems Service Central Bureau.

Montenbruck, O. and Ramos-Bosch, P. (2007). Precise real-time navigation of LEO satellites using GPS measurements. In *GPS Solutions*. Springer-Verlag.

Montenbruck, O., van Helleputte, T., Kroes, R., and Gill, E. (2005). Reduced dynamic orbit determination using GPS code and carrier measurements. In *Aerospace Science and Technology (AST)*. Elsevier, 261-271.

Picone, J.M., Hedin, A.E., Drob, D.P., and Aikin, A.C. (2002). NRLMSISE-00 empirical model of the atmosphere: Statistical comparisons and scientific issues. In *Journal of Geophysical Research*. 107:Issue A12.

Sickle, J.V. (2021). The navigation solution. https://www.e-education.psu.edu/geog862/node/1724. [Online; accessed 15-January-2022].

Swatschina, P. (2012). Dynamic and reduced-dynamic POD of satellites in low earth orbits. In *Department of Geodesy and Geoinformation*. Vienna University.

Weinbach, U. and Schon, S. (2012). Improved GPS receiver clock modelling for kinematic orbit determination of the GRACE satellites. In *IEEE*. 978-1-4673-1923-2/12.

Yang, Y., Yue, X., and Yuan, J. (2015). GPS based reduced-dynamic orbit determination for low earth orbiters with ambiguity fixing. In *International Journal of Aerospace Engineering*. Article ID 723414.